

Module3: CloudWatch Alarms

Assignment

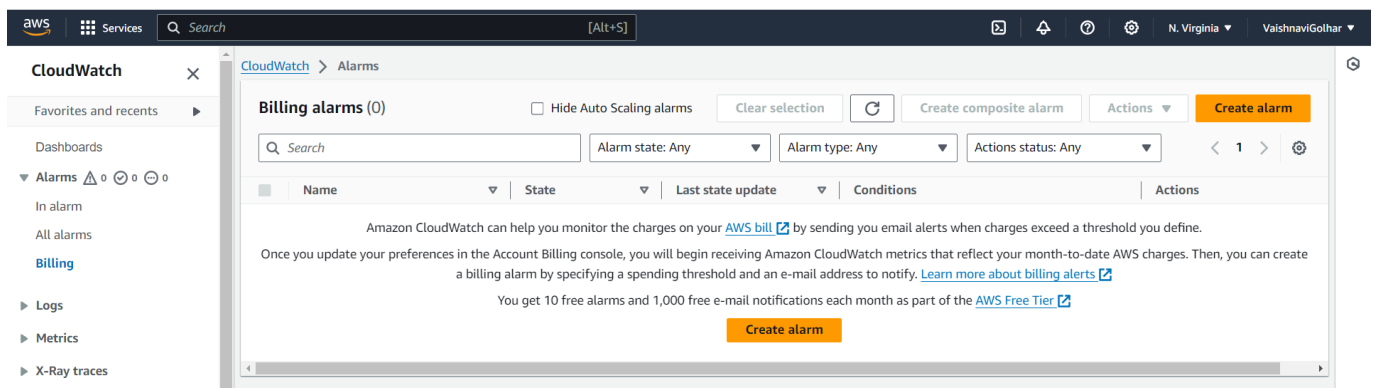
Problem Statement: You work for XYZ Corporation. To maintain the security of the AWS account and the resources you have been asked to implement a solution that can help easily recognize and monitor the different users. Also, you will be monitoring the machines created by these users for any errors or misconfigurations.

Tasks To Be Performed:

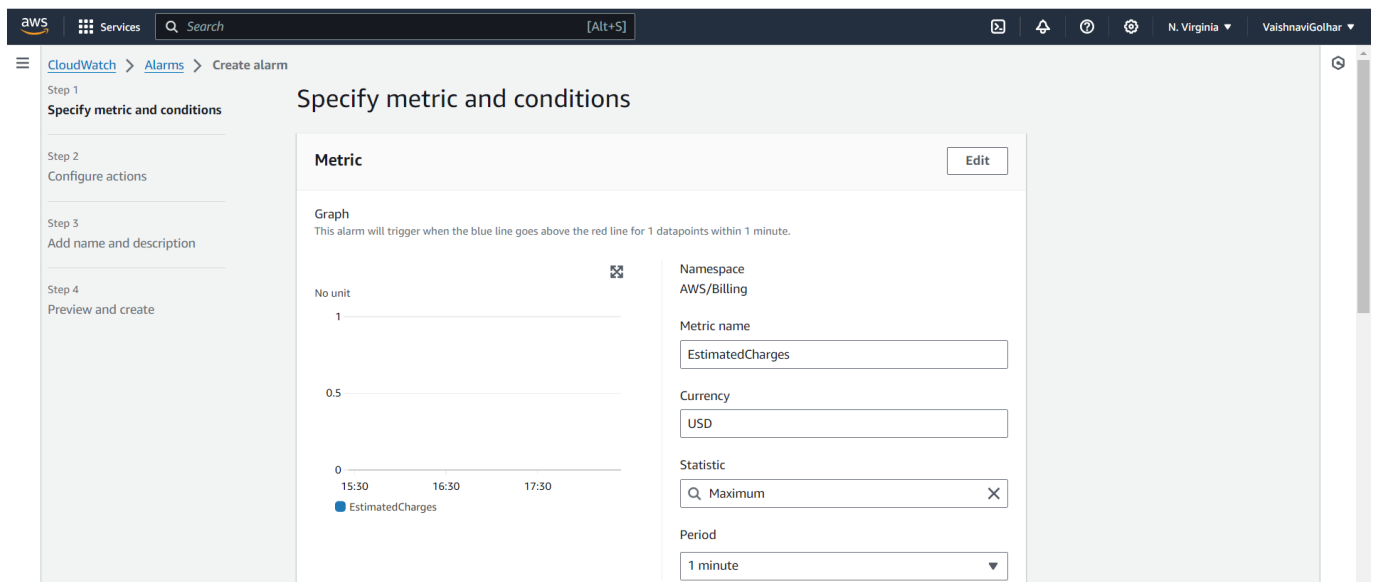
- 1. Create a CloudWatch billing alarm which goes off when the estimated charges go above \$500.**
- 2. Create a CloudWatch alarm which goes off to an Alarm state when the CPU utilization of an EC2 instance goes above 65%. Also add an SNS topic so that it notifies the person when the threshold is crossed.**

Solution: TASK1

1. Log in to AWS Console.
2. Once logged in, navigate to the CloudWatch service.
3. In the CloudWatch dashboard, navigate to the left-hand menu and click on "Billing" under the "Dashboards" section. This will take you to the AWS Billing Dashboard.



4. Specify the metric and conditions accordingly



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"Statistic" to Maximum and set alarm to trigger on 1 consecutive period.

Conditions

Threshold type

☒ Static
Use a value as a threshold

☐ Anomaly detection
Use a band as a threshold

Whenever EstimatedCharges is...

Define the alarm condition.

☒ Greater
> threshold

☐ Greater/Equal
>= threshold

☐ Lower/Equal
<= threshold

☐ Lower
< threshold

than...

Define the threshold value.

500 USD

Must be a number

► Additional configuration

Cancel Next

5. Give your alarm name and description so you can easily identify it later.

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► Additional configuration

Step 2: Configure actions

Edit

Actions

No actions

You don't have any actions for this alarm.

Step 3: Add name and description

Edit

Name and description

Name

Billing-Alarm500\$

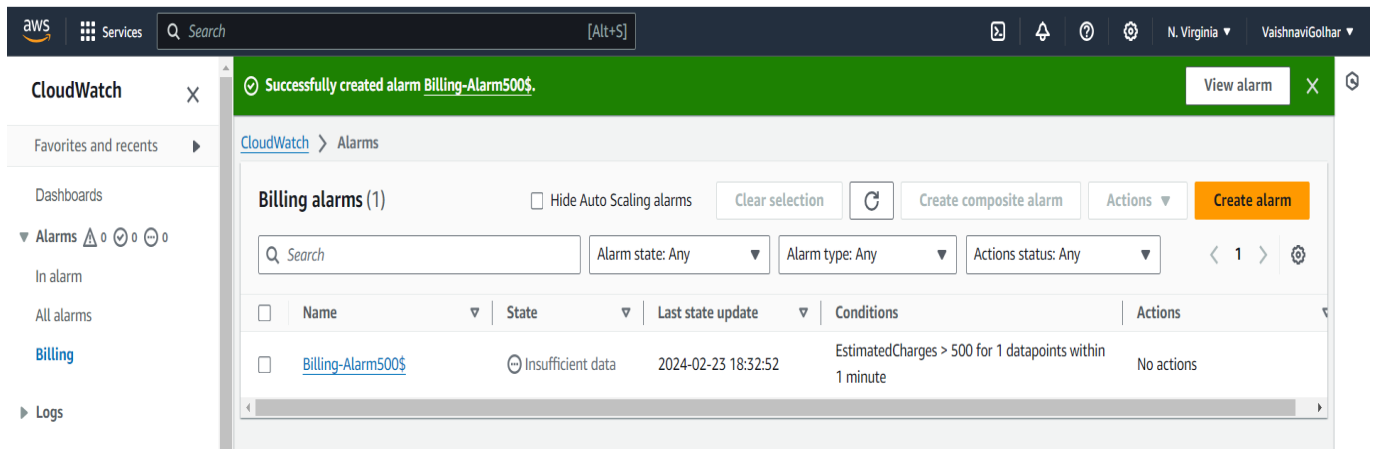
Description

-

Cancel Previous Create alarm

6. Review and Create: Review the alarm configuration to ensure everything is set up correctly, then click "Create Alarm".

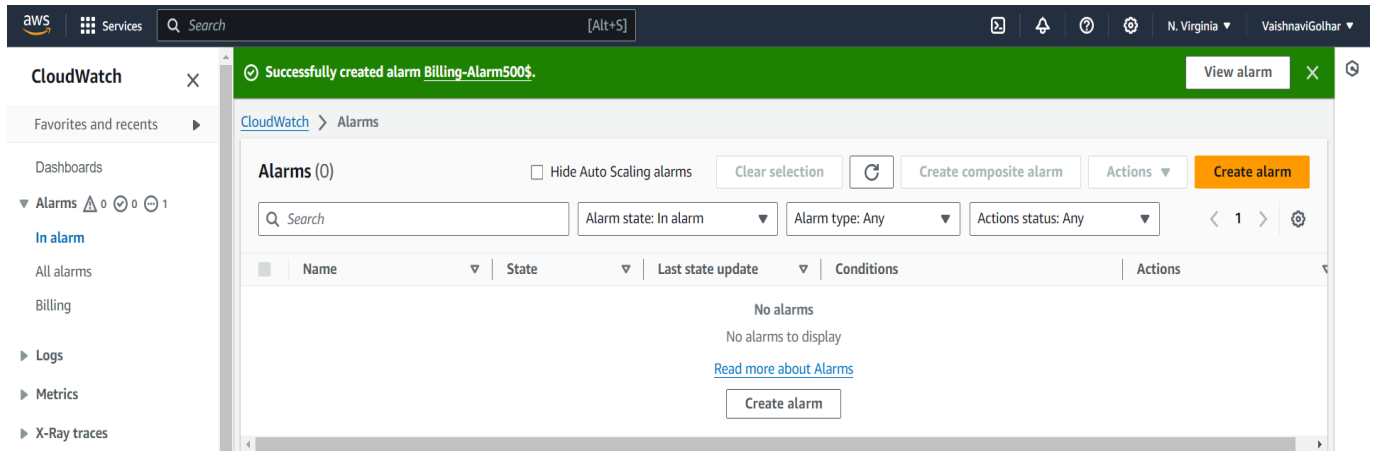
And Alarm has been created.



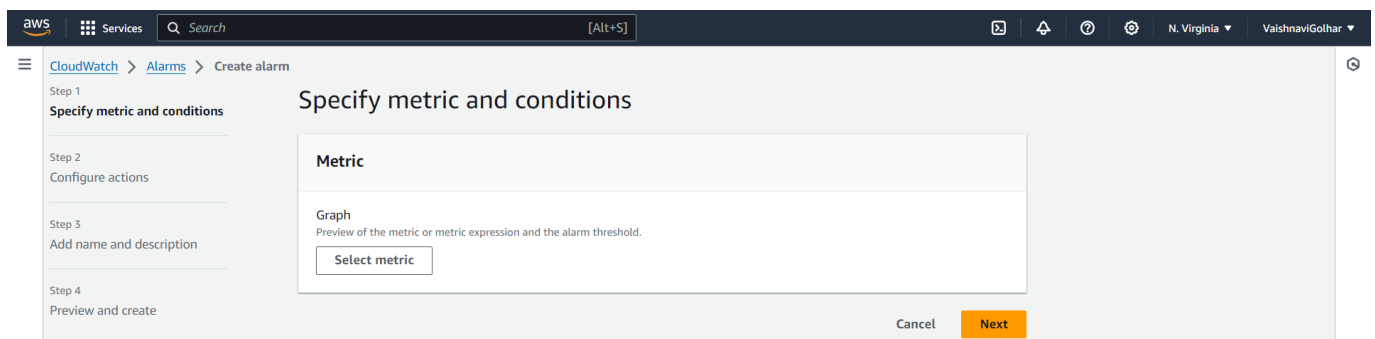
Solution: TASK2

1. Go to the AWS Management Console and navigate to the CloudWatch service.

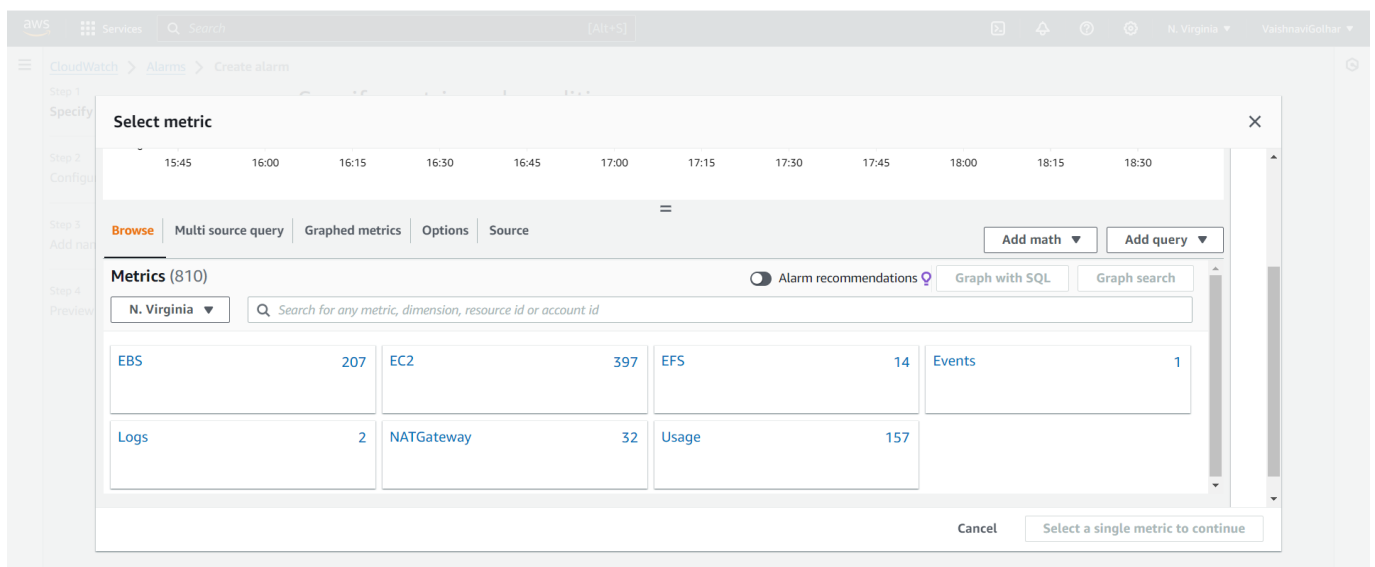
2. In the left-hand navigation pane, click on "Alarms" and then click on the "Create alarm" button.



3. Specify metric and conditions.

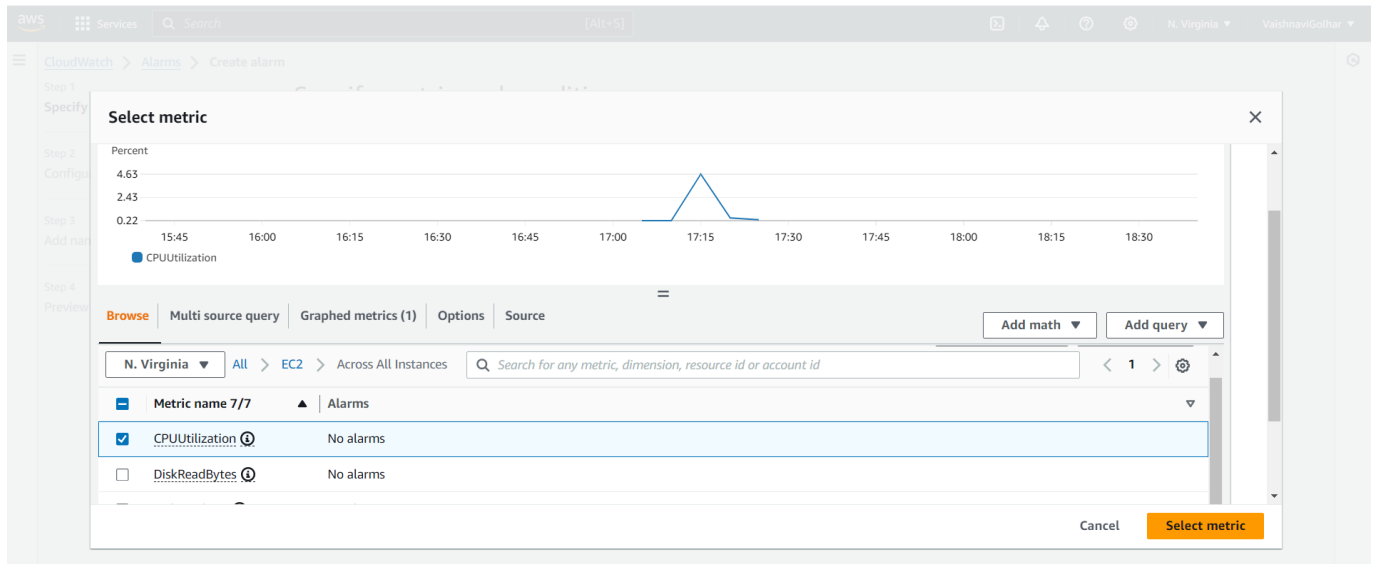


4. Under "All metrics", select "EC2".

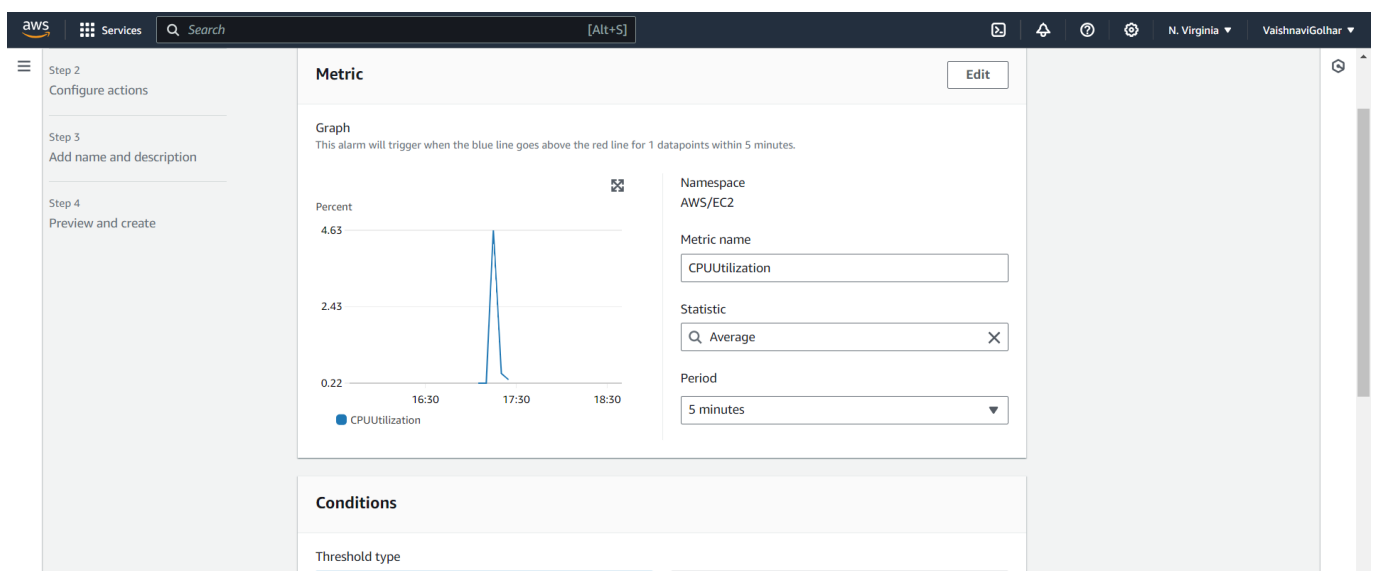


5. Select the metric "Per-Instance Metrics".

Choose the instance for which you want to monitor CPU utilization.



6. Select the "Average" statistic, set the period to 5 minutes, and specify a threshold of 65.



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CPUUtilization 5 minutes

Conditions

Threshold type

☒ Static
Use a value as a threshold

☐ Anomaly detection
Use a band as a threshold

Whenever CPUUtilization is...
Define the alarm condition.

☒ Greater
> threshold

☐ Greater/Equal
>= threshold

☐ Lower/Equal
≤ threshold

☐ Lower
< threshold

than...
Define the threshold value.

65

Must be a number

► Additional configuration

Cancel Next

7. In the Configure actions, Create SNS topic with name.

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CloudWatch > Alarms > Create alarm

Step 1
Specify metric and conditions

Step 2
Configure actions

Step 3
Add name and description

Step 4
Preview and create

Configure actions

Notification

Alarm state trigger
Define the alarm state that will trigger this action.

☒ In alarm
The metric or expression is outside of the defined threshold.

☐ OK
The metric or expression is within the defined threshold.

☐ Insufficient data
The alarm has just started or not enough data is available.

Remove

Send a notification to the following SNS topic
Define the SNS (Simple Notification Service) topic that will receive the notification.

☐ Select an existing SNS topic

☒ Create new topic

☐ Use topic ARN to notify other accounts

Create a new topic...
The topic name must be unique.

Default_CloudWatch_Alarms_Topic

SNS topic names can contain only alphanumeric characters, hyphens (-) and underscores (_).

Email endpoints that will receive the notification...
Add a comma-separated list of email addresses. Each address will be added as a subscription to the topic above.

vaishnavigolhar05@gmail.com

8. Confirm Pending Subscription.

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Amazon SNS

- Dashboard
- Topics
- Subscriptions
- ▼ Mobile
 - Push notifications
 - Text messaging (SMS)
 - Origination numbers

Details

Name	Default_CloudWatch_Alarms_Topic	Display name	-
ARN	arn:aws:sns:us-east-1:161681788148:Default_CloudWatch_Alarms_Topic	Topic owner	161681788148
Type	Standard		

< Subscriptions Access policy Data protection policy Delivery policy (HTTP/S) Delivery status logging Encryption Tags >

Subscriptions (1) Edit Delete Request confirmation Confirm subscription Create subscription

Search

ID	Endpoint	Status	Protocol
Pending confirmation	vaishnavigolhar05@gmail.com	Pending confirmation	EMAIL

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Amazon SNS

- Dashboard
- Topics
- Subscriptions
- ▼ Mobile
 - Push notifications
 - Text messaging (SMS)
 - Origination numbers

New Feature
Amazon SNS now supports in-place message archiving and replay for FIFO topics. [Learn more](#)

Amazon SNS > Topics > Default_CloudWatch_Alarms_Topic

Default_CloudWatch_Alarms_Topic Edit Delete Publish message

Details

Name	Default_CloudWatch_Alarms_Topic	Display name	-
ARN	arn:aws:sns:us-east-1:161681788148:Default_CloudWatch_Alarms_Topic	Topic owner	161681788148
Type	Standard		

Subscriptions Access policy Data protection policy Delivery policy (HTTP/S) Delivery status logging Encryption Tags Integrations

Subscriptions (1) Edit Delete Request confirmation Confirm subscription Create subscription

Search

ID	Endpoint	Status	Protocol
001e99d4-1841-44b7-9431-e09...	vaishnavigolhar05@gmail.com	Confirmed	EMAIL

9. Add name and description to the Alarm.

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CloudWatch

Alarms

Create alarm

Step 1

Specify metric and conditions

Step 2

Configure actions

Step 3

Add name and description

Step 4

Preview and create

Add name and description

Name and description

Alarm name

AlarmAssia4

Alarm description - optional [View formatting guidelines](#)

Edit

Preview

This is an H1
double asterisks will produce strong character
This is [an example](https://example.com/) inline link.

Up to 1024 characters (0/1024)

ⓘ Markdown formatting is only applied when viewing your alarm in the console. The description will remain in plain text in the alarm notifications.

Cancel

Previous

Next

10. Preview and create Alarm

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CloudWatch

Alarms

Alarms (2)

☐ Hide Auto Scaling alarms

Clear selection

Create composite alarm

Actions

Create alarm

Q Search

Alarm state: Any

Alarm type: Any

Actions status: Any

< 1 >

☐	Name	State	Last state update	Conditions	Actions
☐	AlarmAssia4	⊖ Insufficient data	2024-02-23 18:56:56	CPUUtilization > 65 for 1 datapoints within 5 minutes	✅ Actions enabled
☐	Billing-Alarm500\$	⊖ Insufficient data	2024-02-23 18:32:52	EstimatedCharges > 500 for 1 datapoints within 1 minute	No actions